UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 23

Fall 2025

Instructor: Somayeh Sojoudi

Typeset by: George Ma

© EECS 127 - EECS 227AT. Optimization Models in Engineering

Consider the following (convex or non-convex) optimization problem (called *primal problem*):

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \le 0, \ i = 1, \dots, m, \\ & h_j(\boldsymbol{x}) = 0, \ j = 1, \dots, k \end{aligned} \tag{P}$$

and its corresponding *dual problem*:

$$\begin{aligned} \max_{\boldsymbol{\lambda}\in\mathbb{R}^m, \boldsymbol{\mu}\in\mathbb{R}^k} \quad & d(\boldsymbol{\lambda}, \boldsymbol{\mu}), \\ \text{s.t.} \quad & \boldsymbol{\lambda} \ge \mathbf{0}, \end{aligned} \tag{D}$$

where $d(\boldsymbol{\lambda}, \boldsymbol{\mu})$ is the dual function defined in Lecture Note 22.

Theorem 1. *Let the optimal objective values of* (P) *and* (D) *be denoted as* p^* *and* d^**. Then,*

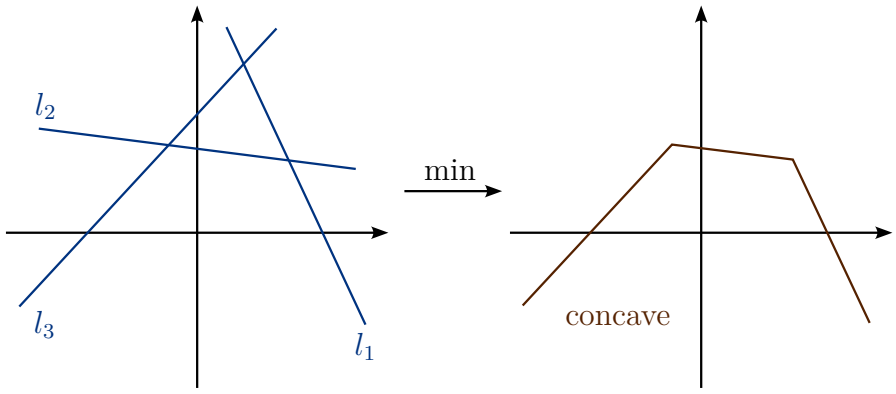
(i) $d(\boldsymbol{\lambda}, \boldsymbol{\mu})$ *is a concave function.*

(ii) $d^* \le p^*$.

- (ii) is called *weak duality* and we proved it earlier.
- (i) is due to the fact that

$$\begin{aligned} d(\boldsymbol{\lambda}, \boldsymbol{\mu}) &= \min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \\ &= \min_{\boldsymbol{x}} \underbrace{\left(f_0(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i f_i(\boldsymbol{x}) + \sum_{j=1}^{k} \mu_j h_j(\boldsymbol{x}) \right)}_{\text{affine in } \lambda_i, \mu_j}. \end{aligned}$$

  There are infinitely many choices for $\boldsymbol{x}$, each gives an affine function of $(\boldsymbol{\lambda}, \boldsymbol{\mu})$, and the (pointwise) minimum of affine functions is concave.



Note:

$$\text{(D)} \iff \begin{aligned} \min \quad & \underbrace{-d(\boldsymbol{\lambda}, \boldsymbol{\mu})}_{\text{convex}}, \\ \text{s.t.} \quad & \underbrace{\boldsymbol{\lambda} \ge \mathbf{0}}_{\text{linear}}. \end{aligned}$$

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.

- While (P) could be convex or non-convex, (D) is always convex.
- So, through convex optimization, we obtain a lower bound on a hard problem.
- If $p^* = d^*$, it is said that the *duality gap* $p^* - d^*$ is zero or *strong duality* holds.

Theorem 2. *If (P) is convex and Slater's condition holds, then strong duality holds.*

Theorem 3. *Assume (P) is convex and Slater's condition holds.*

(i) If $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ satisfies the KKT conditions for (P), then $\boldsymbol{x}^*$ is an optimal solution to (P) and $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ is an optimal solution to (D).

(ii) If $\boldsymbol{x}^*$ is an arbitrary optimal solution to (P) and $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ is an arbitrary optimal solution to (D), then $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ satisfies the KKT conditions for (P).

Under Theorem 3's assumptions, the KKT conditions

(1) Primal feasibility: $f_i(\boldsymbol{x}^*) \le 0,\ i = 1, \dots, m;\ h_j(\boldsymbol{x}^*) = 0,\ j = 1, \dots, k;$

(2) Dual feasibility: $\lambda_i^* \ge 0,\ i = 1, \dots, m;$

(3) Complementary slackness: $\lambda_i^* f_i^*(\boldsymbol{x}^*) = 0,\ i = 1, \dots, m;$

(4) Stationarity: $\nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)|_{\boldsymbol{x}=\boldsymbol{x}^*} = \mathbf{0};$

solve for (P) and (D) simultaneously.

Example 1. Consider the dual of the minimum norm solution for a system of equations:

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^n} \quad & \|\boldsymbol{x}\|_2^2, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}, \end{aligned}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ has full row rank. It can be written equivalently as

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^n} \quad & \boldsymbol{x}^{\mathrm{T}} \boldsymbol{x}, \\ \text{s.t.} \quad & \boldsymbol{a}_i^{\mathrm{T}} \boldsymbol{x} = b_i,\ i = 1, \dots, m. \end{aligned}$$

Let $\boldsymbol{\mu} \in \mathbb{R}^m$ be the Lagrange multiplier. The Lagrangian is

$$L(\boldsymbol{x}, \boldsymbol{\mu}) = \boldsymbol{x}^{\mathrm{T}} \boldsymbol{x} + \sum_{i=1}^{m} \mu_i \left(\boldsymbol{a}_i^{\mathrm{T}} \boldsymbol{x} - b_i \right).$$

- For convex optimization, $L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ is always convex in $\boldsymbol{x}$.
- The minimum of $L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ in $\boldsymbol{x}$ may be finite or $-\infty$. The solution to $\min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ exists iff we can find a point $\boldsymbol{x}^*(\boldsymbol{\lambda}, \boldsymbol{\mu})$ (depends on $\boldsymbol{\lambda}, \boldsymbol{\mu}$) such that

  $$\nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})|_{\boldsymbol{x}=\boldsymbol{x}^*(\boldsymbol{\lambda}, \boldsymbol{\mu})} = \mathbf{0},$$

  i.e., we can find a stationary point of $L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$.

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.

Going back to the example. In our problem,

$$\nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\mu}) = 2\boldsymbol{x} + \underbrace{\mu_1 \boldsymbol{a}_1 + \mu_2 \boldsymbol{a}_2 + \cdots + \mu_m \boldsymbol{a}_m}_{\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu}}.$$

We have $\boldsymbol{x}^* = -\frac{1}{2}\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu}$ and

$$\begin{aligned} d(\boldsymbol{\mu}) &= \min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\mu}) = L\big(\boldsymbol{x}^*(\boldsymbol{\mu}), \boldsymbol{\mu}\big) \\ &= L\left(-\frac{1}{2}\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu}, \boldsymbol{\mu}\right) = -\frac{1}{4}\boldsymbol{\mu}^{\mathrm{T}}\mathbf{A}\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu} - \boldsymbol{\mu}^{\mathrm{T}}\boldsymbol{b}. \end{aligned}$$

Note that $d(\boldsymbol{\mu})$ is concave since $-\mathbf{A}\mathbf{A}^{\mathrm{T}} \preceq \mathbf{0}$. The dual problem is

$$\max_{\boldsymbol{\mu} \in \mathbb{R}^m} \quad -\frac{1}{4}\boldsymbol{\mu}^{\mathrm{T}}\mathbf{A}\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu} - \boldsymbol{\mu}^{\mathrm{T}}\boldsymbol{b}.$$

To maximize, take gradient with respect to $\boldsymbol{\mu}$ and set it to zero:

$$\begin{aligned} \nabla_{\boldsymbol{\mu}} \left(-\frac{1}{4}\boldsymbol{\mu}^{\mathrm{T}}\mathbf{A}\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu} - \boldsymbol{\mu}^{\mathrm{T}}\boldsymbol{b}\right) = \mathbf{0} &\Rightarrow -\frac{1}{2}\mathbf{A}\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu}^* - \boldsymbol{b} = \mathbf{0} \\ &\Rightarrow \boldsymbol{\mu}^* = -2\left(\mathbf{A}\mathbf{A}^{\mathrm{T}}\right)^{-1}\boldsymbol{b}. \end{aligned}$$

In Example 1, Lecture Note 22, we solved KKT for this problem. You can compare and see this dual solution is the same as the KKT solution, which verifies Theorem 3. ■

Theorem 4. *The dual of LP is another LP.*

Proof. Consider the LP:

$$\begin{aligned} \min \quad & \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{x}, \\ \text{s.t.} \quad & \boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{x} = b_i, \ i = 1, \ldots, m, \\ & \boldsymbol{c}_j^{\mathrm{T}}\boldsymbol{x} \le d_j, \ j = 1, \ldots, k. \end{aligned}$$

Let $\boldsymbol{\lambda} \in \mathbb{R}^k, \boldsymbol{\mu} \in \mathbb{R}^m$ be the Lagrange multipliers. The Lagrangian is

$$\begin{aligned} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) &= \boldsymbol{a}_0^{\mathrm{T}}\boldsymbol{x} + \sum_{i=1}^m \mu_i(\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{x} - b_i) + \sum_{j=1}^k \lambda_j(\boldsymbol{c}_j^{\mathrm{T}}\boldsymbol{x} - d_j) \\ &= \underbrace{\boxed{\left(\boldsymbol{a}_0^{\mathrm{T}} + \sum_{i=1}^m \mu_i \boldsymbol{a}_i^{\mathrm{T}} + \sum_{j=1}^k \lambda_j \boldsymbol{c}_j^{\mathrm{T}}\right)\boldsymbol{x}}}_{\text{1st term}} + \underbrace{\boxed{\left(-\sum_{i=1}^m \mu_i b_i - \sum_{j=1}^k \lambda_j d_j\right)}}_{\text{2nd term}}. \end{aligned}$$

- The 2nd term has no $\boldsymbol{x}$ in it.
- The 1st term is in the form of $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}$.
- To minimize $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}$, we set the derivative to zero, i.e., $\boldsymbol{w} = \mathbf{0}$.

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.

- Thus, if $\boldsymbol{a}_0 + \sum \mu_i \boldsymbol{a}_i + \sum \lambda_j \boldsymbol{c}_j = \boldsymbol{0}$, the minimum of $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}$ is zero; and if it is non-zero due to our choices of $(\boldsymbol{\lambda}, \boldsymbol{\mu})$, then there's no stationary point and the minimum of $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}$ is $-\infty$.
- Since $-\infty$ is a useless lower-bound, we impose a constraint to avoid it:

$$\min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \begin{cases} -\sum \mu_i b_i - \sum \lambda_j d_j, & \text{if } \boldsymbol{a}_0 + \sum \mu_i \boldsymbol{a}_i + \sum \lambda_j \boldsymbol{c}_j = \boldsymbol{0}, \\ -\infty, & \text{otherwise.} \end{cases}$$

The dual problem is:

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & -\sum_{i=1}^{m} \mu_i b_i - \sum_{j=1}^{k} \lambda_j d_j, \\ \text{s.t.} \quad & \lambda_i \geq 0, \\ & \boxed{\boldsymbol{a}_0 + \sum_{i=1}^{m} \mu_i \boldsymbol{a}_i + \sum_{j=1}^{k} \lambda_j \boldsymbol{c}_j = \boldsymbol{0}}. \end{aligned}$$

- The dual problem is LP in $\boldsymbol{\lambda}, \boldsymbol{\mu}$.
- Since LP has no non-affine constraints, Slater's condition means there exists $\boldsymbol{y}$ such that $f_i(\boldsymbol{y}) \leq 0$, $h_j(\boldsymbol{y}) = 0$. This is equivalent to feasibility. Therefore, the dual of LP is LP, and the duality gap is zero if the LP is feasible.

■

Theorem 5. *The dual of QP is another QP.*

Proof. Consider the QP:

$$\begin{aligned} \min \quad & \boldsymbol{x}^{\mathrm{T}}\mathbf{P}_0\boldsymbol{x} + \boldsymbol{q}_0^{\mathrm{T}}\boldsymbol{x} + r_0, \\ \text{s.t.} \quad & \boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{x} = b_i, \ i = 1, \ldots, m, \\ & \boldsymbol{c}_j^{\mathrm{T}}\boldsymbol{x} \leq d_j, \ j = 1, \ldots, k, \end{aligned}$$

where $\mathbf{P}_0 \succeq \mathbf{0}$. Let $\boldsymbol{\lambda} \in \mathbb{R}^k, \boldsymbol{\mu} \in \mathbb{R}^m$ be the Lagrange multipliers. The Lagrangian is

$$\begin{aligned} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) &= \boldsymbol{x}^{\mathrm{T}}\mathbf{P}_0\boldsymbol{x} + \boldsymbol{q}_0^{\mathrm{T}}\boldsymbol{x} + r_0 + \sum_{i=1}^{m} \mu_i(\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{x} - b_i) + \sum_{j=1}^{k} \lambda_j(\boldsymbol{c}_j^{\mathrm{T}}\boldsymbol{x} - d_j) \\ &= \boldsymbol{x}^{\mathrm{T}}\mathbf{P}_0\boldsymbol{x} + \left(\boldsymbol{q}_0^{\mathrm{T}} + \sum_{i=1}^{m} \mu_j \boldsymbol{a}_i^{\mathrm{T}} + \sum_{j=1}^{k} \lambda_j \boldsymbol{c}_j^{\mathrm{T}}\right)\boldsymbol{x} + \left(r_0 - \sum_{i=1}^{m} \mu_i b_i - \sum_{j=1}^{k} \lambda_j d_j\right), \end{aligned}$$

which is convex and quadratic in $\boldsymbol{x}$. To minimize $L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ with respect to $\boldsymbol{x}$, we solve

$$\nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \boldsymbol{0} \ \Rightarrow \ 2\mathbf{P}_0\boldsymbol{x} + \left(\boldsymbol{q}_0 + \sum_{i=1}^{m} \mu_i \boldsymbol{a}_i + \sum_{j=1}^{k} \lambda_j \boldsymbol{c}_j\right) = \boldsymbol{0}.$$

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.

Note: this may not have a solution if for example $\mathbf{P}_0 = \mathbf{0}$ or $\boldsymbol{q}_0 + \sum \mu_i \boldsymbol{a}_i + \sum \mu_j \boldsymbol{c}_j$ is not in the range of $\mathbf{P}_0$. To avoid such scenarios (taking dual is more complicated in that case), assume $\mathbf{P}_0 \succ \mathbf{0}$, then

$$\boldsymbol{x}^*(\boldsymbol{\lambda}, \boldsymbol{\mu}) = -\frac{1}{2}\mathbf{P}_0^{-1}\boldsymbol{g}(\boldsymbol{\lambda}, \boldsymbol{\mu})$$

where $\boldsymbol{g}(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \boldsymbol{q}_0 + \sum \mu_i \boldsymbol{a}_i + \sum \lambda_j \boldsymbol{c}_j$. The dual function is

$$\begin{aligned} d(\boldsymbol{\lambda}, \boldsymbol{\mu}) &= L\big(\boldsymbol{x}^*(\boldsymbol{\lambda}, \boldsymbol{\mu}), \boldsymbol{\lambda}, \boldsymbol{\mu}\big) \\ &= \boldsymbol{x}^*(\boldsymbol{\lambda}, \boldsymbol{\mu})^{\mathrm{T}}\mathbf{P}_0\boldsymbol{x}^*(\boldsymbol{\lambda}, \boldsymbol{\mu}) + \boldsymbol{q}_0^{\mathrm{T}}\boldsymbol{x}^*(\boldsymbol{\lambda}, \boldsymbol{\mu}) + r_0 \\ &\quad + \left(\sum_{i=1}^m \mu_i \boldsymbol{a}_i + \sum_{j=1}^k \lambda_j \boldsymbol{c}_j\right)^{\mathrm{T}} \boldsymbol{x}^*(\boldsymbol{\lambda}, \boldsymbol{\mu}) - \left(\sum_{i=1}^m \mu_i b_i + \sum_{j=1}^k \lambda_j d_j\right) \\ &= -\frac{1}{4}\boldsymbol{g}(\boldsymbol{\lambda}, \boldsymbol{\mu})^{\mathrm{T}}\mathbf{P}_0^{-1}\boldsymbol{g}(\boldsymbol{\lambda}, \boldsymbol{\mu}) + \left(r_0 - \sum_{i=1}^m \mu_i b_i - \sum_{j=1}^k \lambda_j d_j\right). \end{aligned}$$

The dual problem is given by

$$\begin{aligned} \max \quad & -\frac{1}{4}\left(\boldsymbol{q}_0 + \sum_{i=1}^m \mu_i \boldsymbol{a}_i + \sum_{j=1}^k \lambda_j \boldsymbol{c}_j\right)^{\mathrm{T}} \mathbf{P}_0^{-1} \left(\boldsymbol{q}_0 + \sum_{i=1}^m \mu_i \boldsymbol{a}_i + \sum_{j=1}^k \lambda_j \boldsymbol{c}_j\right) \\ & + r_0 - \sum_{i=1}^m \mu_i b_i - \sum_{j=1}^k \lambda_j d_j, \\ \text{s.t.} \quad & \boldsymbol{\lambda} \geq \mathbf{0}. \end{aligned}$$

The objective of the dual problem is quadratic and concave in $(\boldsymbol{\lambda}, \boldsymbol{\mu})$. This is because $\boldsymbol{y}^{\mathrm{T}}\mathbf{P}_0^{-1}\boldsymbol{y}$ is convex in $\boldsymbol{y}$, and if we change $\boldsymbol{y}$ to a linear function $\boldsymbol{g}(\boldsymbol{\lambda}, \boldsymbol{\mu})$ (linear transformation), we have $\boldsymbol{g}(\boldsymbol{\lambda}, \boldsymbol{\mu})^{\mathrm{T}}\mathbf{P}_0^{-1}\boldsymbol{g}(\boldsymbol{\lambda}, \boldsymbol{\mu})$ is also convex in $(\boldsymbol{\lambda}, \boldsymbol{\mu})$. Thus $-\boldsymbol{g}(\boldsymbol{\lambda}, \boldsymbol{\mu})^{\mathrm{T}}\mathbf{P}_0^{-1}\boldsymbol{g}(\boldsymbol{\lambda}, \boldsymbol{\mu})$ is concave in $(\boldsymbol{\lambda}, \boldsymbol{\mu})$, and the dual problem is QP. ■

Consider the following optimization problem:

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \leq 0, \ i = 1, \ldots, k, \\ & h_j(\boldsymbol{x}) = 0, \ j = 1, \ldots, m. \end{aligned}$$

There are three possibilities:

(1) The optimization is unbounded from below (e.g., $\min x$): $p^* = -\infty$.

(2) The solution exists: p^* is finite.

(3) The feasible set is empty: $p^* = +\infty$ by convention (e.g., $\min x$ s.t. $x \leq 0, \ x \geq 1$).

Now consider the dual problem:

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & d(\boldsymbol{\lambda}, \boldsymbol{\mu}), \\ \text{s.t.} \quad & \boldsymbol{\lambda} \geq \mathbf{0}. \end{aligned}$$

There are also three possibilities:

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.

(1) The optimization is unbounded from above (e.g., $\max \lambda$ s.t. $\lambda \geq 0$): $d^* = +\infty$.

(2) The solution exists: d^* is finite.

(3) The problem is infeasible: $d^* = -\infty$ by convention.

Note: As seen in LP, there could be more constraints in (D) other than $\boldsymbol{\lambda} \geq \mathbf{0}$ to make sure $d(\boldsymbol{\lambda}, \boldsymbol{\mu})$ is well-defined. So, a feasible $(\boldsymbol{\lambda}, \boldsymbol{\mu})$ doesn't always exist.

- When we say weak duality implies $p^* \geq d^*$, we allow p^* and d^* to be finite or $\pm\infty$. This is an important fact in proving infeasibility.
- Given a set of constraints: $f_i(\boldsymbol{x}) \leq 0,\ i = 1, \ldots, k$; $h_j(\boldsymbol{x}) = 0,\ j = 1, \ldots, m$; the goal is to certify that there is no $\boldsymbol{x}$ satisfying all constraints. This seems hard since we can't plug in infinitely many choices of $\boldsymbol{x} \in \mathbb{R}^n$ into the constraints to show some are violated.
- In this case, we can use duality.
- Consider the following optimization:

$$\begin{aligned} \min \quad & 0, \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \leq 0,\ i = 1, \ldots, k, \\ & h_j(\boldsymbol{x}) = 0,\ j = 1, \ldots, m. \end{aligned}$$

  So, we minimize 0 over the constraints. There are two choices for p^*:

  - $p^* = 0$ if the set of constraints is feasible and thus the optimization has a solution;
  - $p^* = +\infty$ if the problem is infeasible.

- Consider the dual optimization:

$$\begin{aligned} \max \quad & d(\boldsymbol{\lambda}, \boldsymbol{\mu}), \\ \text{s.t.} \quad & \boldsymbol{\lambda} \geq \mathbf{0}. \end{aligned}$$

- We know that $d^* \geq d(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}})$ if $\hat{\boldsymbol{\lambda}} \geq \mathbf{0}$, since d^* is the maximum.
- If we can find a pair $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}})$ with $\hat{\boldsymbol{\lambda}} \geq \mathbf{0}$ such that $d(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}}) > 0$, then

$$p^* \geq d^* \geq d(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}}) > 0 \ \Rightarrow\ p^* \neq 0 \ \Rightarrow\ p^* = +\infty.$$

  Therefore, finding this single point certifies that

$$\begin{cases} f_i(\boldsymbol{x}) \leq 0,\ i = 1, \ldots, k, \\ h_j(\boldsymbol{x}) = 0,\ j = 1, \ldots, m \end{cases}$$

  has no solution. Such point $(\hat{\boldsymbol{\lambda}}, \hat{\boldsymbol{\mu}})$ is called a *certificate*.

© EECS127/227AT. Optimization Models in Engineering, Somayeh Sojoudi, Fall 2025.